



ROBOTIC SOLUTIONS



DUET

- Friendly
- Expressive
- Interactive Robot

ABOUT DUET

Duet is a friendly interactive expressive mobile robot designed to work as a stand-alone promoter who can welcome the guests, guide them, conduct conversations, check in/out, print & scan documents, and provides information in an attractive way using its changeable facial expressions.

The Robot Duet developed by MARSES Robotic Solutions to play many roles to be the best innovative marketing tool choice; it works in different environments such as (Hotels, Resorts, Banks, Exhibitions, Restaurants, Companies, Airports, Museums, Hospitals Receptions, etc.).



ROLES THAT DUET CAN ACT AS

DIGITAL ASSISTANT BANKING:

These days the Digital Assistant Robots are used by banks and other financial institutions to automate manual business processes so the banks can remain competitive in today's market. We can define the digital assistant robot in banking as the use of a combination of robotics and Artificial Intelligence (AI) technologies to replace and augment human operations in banking.

Banks use The Digital Assistant Robots to perform repetitive tasks like data entry, automate customer service and back-office workflows, print & scan documents and other banking services that can be done without human interface with interaction with the robot through conversation, its interactive touch screen, and facial expressions.

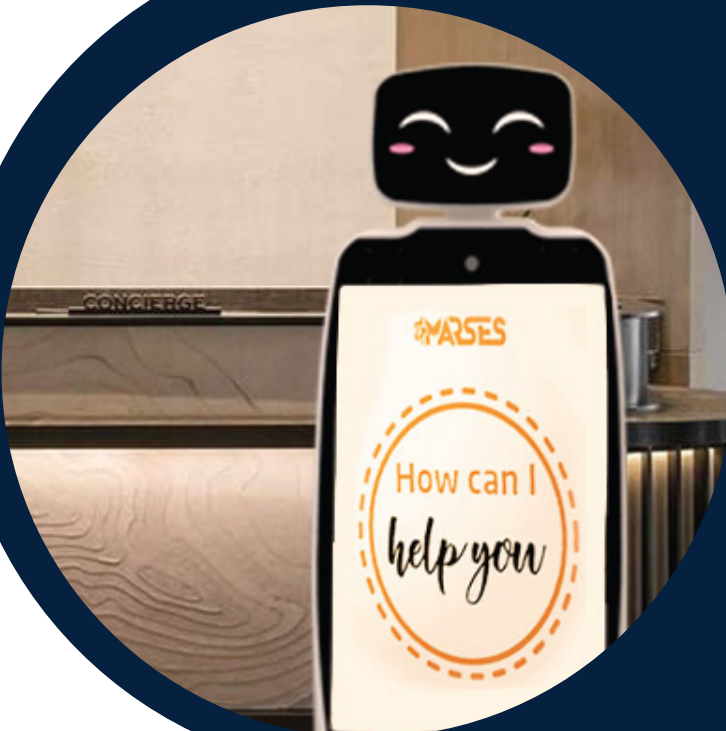


FUNCTIONS



DIGITAL ADMINISTRATOR

If you're looking to reduce your business costs, or simply need a little help to perform your administration tasks smoothly, so you are in a need of MARSES Digital Administration Robot. This new technology can be used to automate your admin tasks, saving valuable time and money and increasing staff satisfaction at the same time. Duet the digital administrator work in an unattended manner, meaning once programmed they complete the task from start to finish.



Duet makes a good first impression to clients, potential customers and visitors. Your reception area is the first port of entry, so it makes sense to ensure you use all the tools at your disposal to create a warm and welcoming space in your reception area, so using Duet for this task absolutely helps with this.

FUNCTIONS



Welcoming & greeting visitors

Open applications for visitors to record their purpose of visit or for Registration

Allow contactless registration
(Scan QR Code to open form on his mobile)

Print visitors passes

Issue an appointment with the employees according to the purpose of visit.

Get insightful reports based on its recorded database

SELF-SERVICE ROBOT

The Self-service Robots have become a frequent texture in many of our lives. Whether you are looking for a quick way to pay and go at supermarket checkouts, ordering at a fast-food outlet, or using information points in train stations, airports, shopping malls, at conferences or events. Duet Robot makes all these functions plus the mobility & interactivity features, which make things easier for customers and businesses and offers the ability to automate simple processes, and reducing the need for additional manpower. Duet with its interactivity features through the interactive facial expressions and interactive touch screen besides the mobility feature which is very important to the guests to get their services or information in a unique, innovative, & interactive way.

FUNCTIONS

- placing orders for food.
- allow payments.
- Print Receipts.
- Display information.
- Facilitate actions.
- Find location & Show interactive maps.



DIGITAL TOUR-GUIDE

When exploring a new place, the activity you think to do is visiting museums, castles, and other historical sites. However, seeing a tourist site without the aid of a tourist guide, some would feel lost and confused. MARSES introduces the solution of The Digital Tour-Guide Robot to ease the process of tourist guides.

The robot will replace the current human guide. The robot will follow the tourists wherever they go, avoiding obstacles in its way, and provide information about the place once the tourists have given it the command to do so.

FUNCTIONS

- Welcoming & greeting visitors.
- Provide historical information.
- physically guide the visitors.
- Display documentaries.
- Show interactive maps.
- Guide in multi languages.
- Answer Related Questions based on its database.



DIGITAL CINEMA BOOKING:

Numerous people are constantly looking at movie theatres and cinemas for entertainment daily. However, these customers' excitement can be crushed by the long lines involved in the manual process of booking and buying tickets. What if These processes are done by an interactive robot that welcomes and interacts with guests? The Digital Booking Robot allows customers to access the services of a business, reserve seats, and buy tickets. The robot-customized software provides details such as when a movie will be played, what seats are available, movie previews, and so much more.



FUNCTIONS

- Book Ticket.
- Print Tickets.
- Allow payments.
- Book seats.
- Display movies promotions.
- Allow Feedback & Reviews.
- Innovative marketing tool.

MAIN FEATURES OF DUET



INTERACTIVE & ATTRACTIVE ROBOT

DUET was created to communicate with people by using his Voice & Facial Expressions to communicate with people. Its main Touch Screen makes the robot interact more with people, enabling it to provide more information attractively.

SELF-SERVICE ROBOT

Duet is characterized by its changeable facial expressions, which make it expressive enough to interact with people. Duet interacts emotionally with people according to their behaviors & orders. Thanks to its facial expressions & voice intonations, Duet creates an empathic relationship between guests.





INTERACTIVE TOUCH SCREEN

Having a displaying 27-inch touch screen makes it shows information in an interactive & innovative way. It shows various promo or informative videos to market your brand & show your ads through a mobile interactive Robot.

USER FRIENDLY

Thanks to its design, size, and humanoid features, everyone can easily accept and communicate with Duet besides its easy-to-use applications & Software

NAVIGATION & LOCALIZATION WITH HIGH ACCURACY

Duet autonomously navigates to find the most efficient path to its destinations. With real-time positioning, the robot adjusts the path when it encounters obstacles, not on the map.

SAFE MANEUVERS IN A POPULATED WORKSPACE

The robot is designed to operate among people and maneuvers safely & efficiently in even highly dynamic environments.

SORTING DATA

The robot can collect data from your (Clients / Guests / Visitors) through its interactive touch screen, then store it to enable you to use it the way you want.

DECELERATION FOR OBJECTS

The built-in sensors ensure that the robot is slowed down when obstacles are detected in front of it.



TECHNICAL SPECIFICATIONS

WEIGHT (WITHOUT LOAD): **35 KG**

GROUND CLEARANCE: **50 MM / 2 IN**

COLOR: **RAL COLOR RAL 9010 / PURE WHITE**

SPEED AND PERFORMANCE

Battery running time: Up to 6 Hours

Maximum speed: 0.5 m/s

Cruise Speed: 0.3-0.5 m/s Adjustable

POWER

Battery Capacity: DC 24V

Battery Type: Lithium

Battery Working Hours: Up to 6 Hours

Type of Charging: Manual / Auto Modes

Battery Charging Time:

- With cable: up to 1.5 Hours

- With charging station: up to 3 Hours

Battery Charge Cycles: 600 - 800 cycle

External charger Input: 100-230 V ac, 50-60 Hz

Output: 24 V, max 15 A

Over Charging Cut o : Available

Over Heat Detection: Available



SOFTWARE

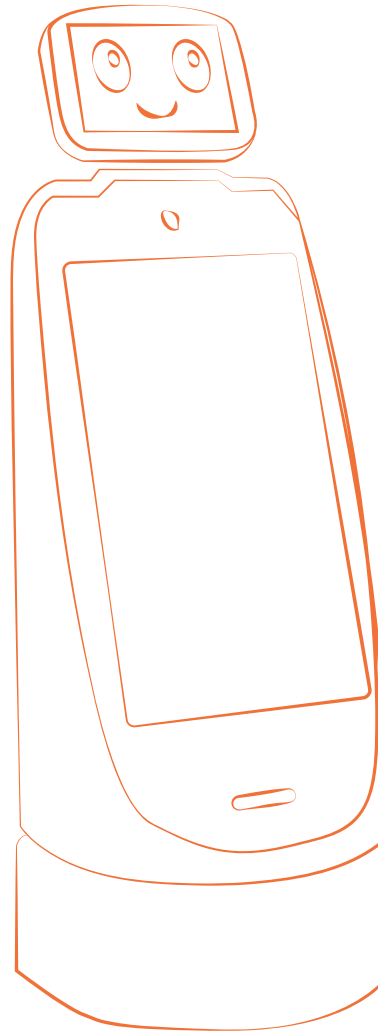
Operating system: Windows

ENVIRONMENT

Ambient temperature range: +5°C to 45°C
(humidity 10-95% non-condensing)
Environment: Indoor, Smooth Floor
Ground clearance: 30 mm

COMMUNICATION

WiFi: Dual-band wireless AC/G/N/
Bluetooth: 4.0 LE
I/Os: USB port



SENSOR SYSTEM

Laser Range Scanner: Range: 360°

RGB-D camera: Detection of objects within 10m range

Object detector sensor: Range 20 cm Min., 2000 cm Max.

RGB-D CAMERA

Camera positioned on the front of the robot detects objects within 10m range.

OBJECT DETECTOR SENSORS

Two sensors are placed on the robot: one in each side and used to detect transparent objects.

ROBOT INTERACTION

Screen: Touch Screen available for variable sizes (Medium & Large) to fit its multi uses

- Face Screen to display its interactive expressions

sound: Chat-bot available

lighting: Attractive lights to Indicate the robot status



There is a correspondence between the robot lights, sound and expressions interactions. Duet changes his expressions and lights code with the efficient sounds according to the current situation that he intelligently detects in real time sense.



SAFETY LASER SCANNERS:

Safety laser scanner has a 360° field of view with 18m/12m Range Radius around the robot, providing visual protection around the robot

SAFETY:

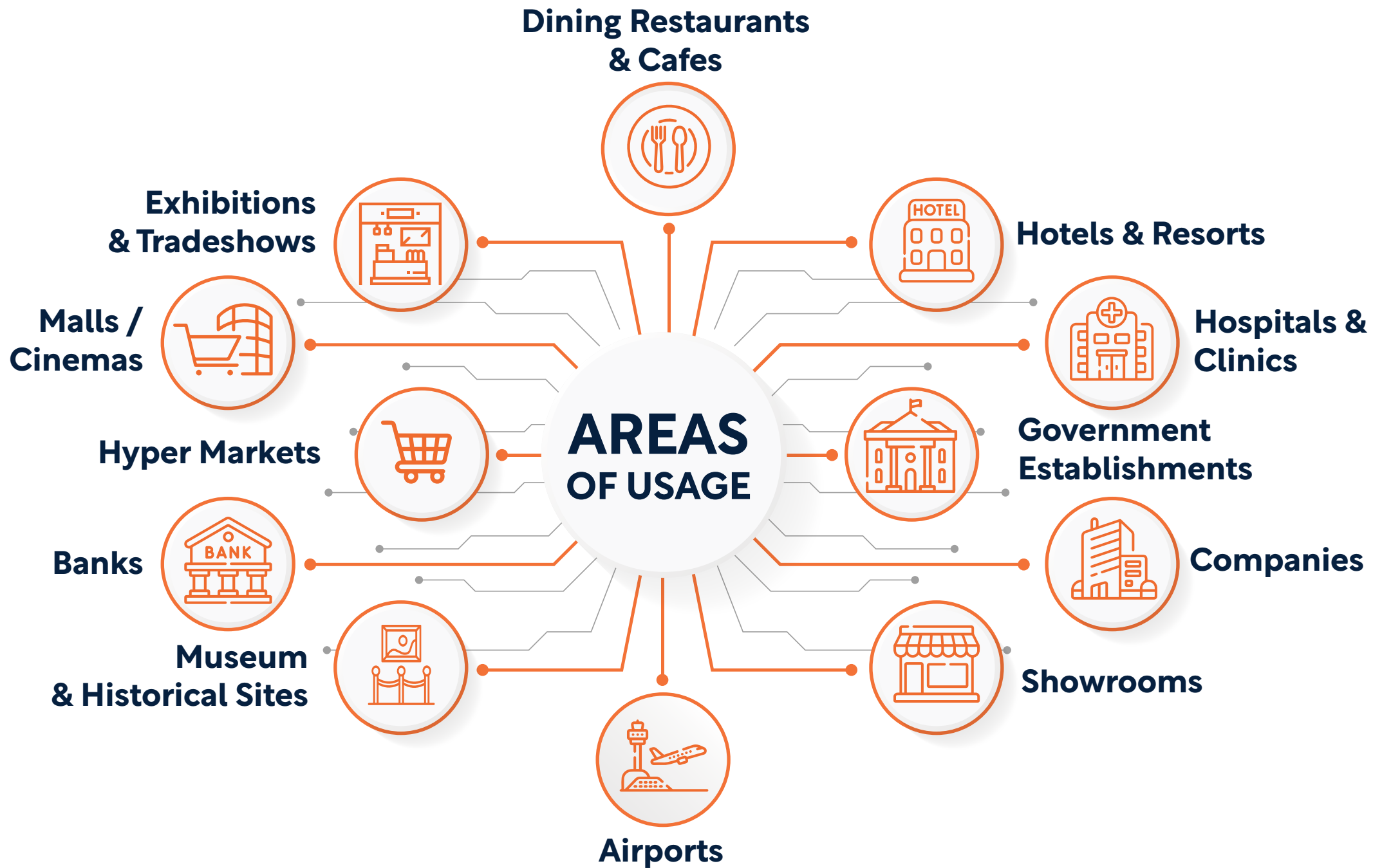
- Emergency Button: Physical Safety System

COLLISION AVOIDANCE:

The collision avoidance safety function ensures that the robot will come to a stop before it collides with a human or object with its 3D intelligent obstacle avoidance technology. In addition, it is automatically deactivated two seconds after the protective field is free.

OVER SPEED AVOIDANCE:

- The safety system monitors if the speed of each motor is above limits for maximum rated speed and hence an indication of speed control is lost for any reason



THANK YOU

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